$$CH_3COOH + OH^-(aq) \rightarrow CH_3COO^-(aq) + H_2O\ (l)$$

$\therefore [CH_3COOH] = 0.8 - \alpha - 0.01 = 0.79 - \alpha$

$[CH_3COO^-] = \alpha + 0.8 + 0.01 = 0.81 + \alpha$ $\quad \alpha << 0.8;$

$0.79 - \alpha \simeq 0.79$ and $0.81 + \alpha \simeq 0.81$

$$\therefore [H^+] = (1.8 \times 10^{-5}) \times \frac{0.79}{0.81}$$

$$[H^+] = 1.76 \times 10^{-5}$$

$$\therefore pH = -\log (1.76 \times 10^{-5})$$

$$= 5 - \log 1.76$$

$$= 5 - 0.25$$

$$pH = 4.75$$

The addition of a strong base (0.01 M NaOH) increased the pH only slightly ie., from 4.74 to 4.75 . So, the buffer action is verified.

Evaluate yourself - 8

a) Explain the buffer action in a basic buffer containing equimolar ammonium hydroxide and ammonium chloride.

b) Calculate the pH of a buffer solution consisting of 0.4M CH_3COOH and 0.4M CH_3COONa . What is the change in the pH after adding 0.01 mol of HCl to 500ml of the above buffer solution. Assume that the addition of HCl causes negligible change in the volume. Given: ($K_a = 1.8 \times 10^{-5}$.)

8.7.2 Buffer capacity and buffer index

The buffering ability of a solution can be measured in terms of buffer capacity. Vanslyke introduced a quantity called buffer index, β, as a quantitative measure of the buffer capacity. It is defined as the number of gram equivalents of acid or base added to 1 litre of the buffer solution to change its pH by unity.

$$\beta = \frac{dB}{d(pH)} \qquad \text{....(8.19)}$$

Here,

dB = number of gram equivalents of acid / base added to one litre of buffer solution.

d(pH) = The change in the pH after the addition of acid / base.

8.7.3 Henderson – Hasselbalch equation

We have already learnt that the concentration of hydronium ion in an acidic buffer solution depends on the ratio of the concentration of the weak acid to the concentration of its conjugate base present in the solution i.e.,

$$[H_3O^+] = K_a \frac{[acid]_{eq}}{[base]_{eq}} \qquad \text{....(8.20)}$$

XII U8-Ionic equilibrium.indd 18 16-12-2022 19:34:03